

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2025

**CM1 - Actuarial Mathematics for Modelling
(Paper A)**

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Kindly note: Do not round off at any intermediate level in any of the question.

Q.1) What is **not** the condition for Redington's immunization?

- (i) The value of the assets at the starting rate of interest is equal to the value of the liabilities.
 - (ii) The volatilities of the asset and liability cashflow series are equal
 - (iii) The convexity of the asset cashflow series is equal to the convexity of the liability cashflow series
-
- a) Only (i)
 - b) Only (ii)
 - c) Only (iii)
 - d) (ii) and (iii)
 - e) None of these

[2]

Q.2) Calculate the expectation of the present value of the benefits from guaranteed annuity issued to a life aged exactly 40, under which payments of INR 10,000 are made annually in arrears for a minimum of 20 years and for life thereafter. Assuming that the annual effective interest rate is 4%, and mortality basis is AM92.

- a) 259,660
- b) 195,945
- c) 196,684
- d) 192,384
- e) None of these

[2]

Q.3) The company offers different type of options to annuitants in terms of choosing the frequency of the annuity payments. For interest rate, $i > 0$, arrange the present value of annuities of the below options in ascending order.

- i. Annuity payable in arrear
 - ii. Annuity payable in advance
 - iii. Annuity payable in arrear in quarterly mode
 - iv. Annuity payable in advance in quarterly mode
 - v. Annuity payable continuously
-
- a) $i < iii < v < iv < ii$
 - b) $i < iii < v < ii < iv$
 - c) $iii < i < v < iv < ii$
 - d) $i < v < iii < iv < ii$
 - e) $iii < i < iv < v < ii$

[2]

Q.4) Mortality table provides the value of A_{50} as 0.3 at an interest rate of 5% p.a. effective. Calculate $\bar{a}_{50}$.

- a) 14.2
- b) 14.7
- c) 13.7
- d) 14.5
- e) None of these

[2]

Q.5) A loan is to be repaid by a series of instalments made annually in arrears. The first instalment is INR 500 per annum and thereafter instalments increase by INR 50 each year. The instalments are paid for 5 years and are calculated using an effective rate of interest of 7% per annum. Calculate the loan amount

- a) 2637
- b) 2603
- c) 2194
- d) 2432
- e) None of these

[2]

For Q6 and Q7- Consider a share that pays a dividend of Rs. 2 annually in arrears in perpetuity at an annual effective interest rate of 5%.

Q.6) Calculate the present value

- a) 42
- b) 40
- c) 38
- d) 45
- e) None of these

[2]

Q.7) Calculate the convexity of the share's cashflows.

- a) 850
- b) 32000
- c) 800
- d) 840
- e) None of these

[2]

Q.8) A company uses an investment return assumption of 6.0% for profit testing. The risk discount rate is same as well. During a profit testing exercise, the company decided to increase the overall prudence margin in reserving from 6% to 8%. The NPV of future profits at the inception of the contracts:

- a) Will increase by 2%
- b) Will decrease by 2%
- c) Will remain unchanged
- d) Will increase by 0.15%

- e) None of these

[2]

Q.9) The force of interest, $\delta(t)$ is a function of time and at any time t (measured in years) is given by

$$\delta(t) = 0.07 - 0.005t \text{ for } t \leq 8$$

$$\delta(t) = 0.06 \text{ for } t > 8$$

Calculate the accumulation at time $t = 10$ of INR 1,000 invested at time $t = 0$.

- a) 1682
- b) 1822
- c) 1162
- d) 1552
- e) None of these

[2]

Q.10) How many days does one need to hold a 360 days Government Bond redeemable at INR 100 if he buys at INR 97.5 and sells at INR 98.5 after achieving a return of 4.5% per annum effective. Consider 360 days in a year. Select the correct option from the choices provided below:

- a) 82 days
- b) 86 days
- c) 85 days
- d) 80 days
- e) None of these

[2]

Q.11) At time $t = 0$, the n -year spot rate of interest is equal to $(2.25 + 0.25n)\%$ per annum effective. Calculate the 2-year forward rate of interest from time $t = 3$ expressed as an annual effective rate of interest.

- a) 8.01%
- b) 4.25%
- c) 8.69%
- d) 4.50%
- e) None of these

[2]

Q.12) A two-year term assurance policy is issued to a life aged 50 exact. The benefit amount is 100,000 and are payable at the end of the year of death. What is the expected present value of the benefits from this policy using PMA92C20 mortality and an effective interest rate of 6%?

- a) 107
- b) 103
- c) 113
- d) 109

- e) None of these

[2]

Q.13) Calculate the value of $4.75 p_{30.75}$ using AM92 Ultimate mortality and assuming that deaths are uniformly distributed between integer ages.

- a) 0.9172
- b) 0.9872
- c) 0.9780
- d) 0.9970
- e) None of these

[3]

Q.14) A population is subject to a constant force of mortality of 0.015. Calculate the curtate expectation of a life aged 20 exact.

- a) 55.12
- b) 65
- c) 66.17
- d) 62
- e) None of these

[2]

Q.15) Calculate $q_{[50]:[60]}$ for two independent lives basis AM92 select.

- a) 0.0077
- b) 0.0038
- c) 0.0058
- d) 0.0105
- e) None of these

[2]

Q.16) Calculate $a_{55|50}$ assuming 55 is male and 50 is female. Mortality table is PMA92C20 for the male life, PFA92C20 for the female life. Interest is 4% effective.

- a) 2.973
- b) 2.937
- c) 1.608
- d) 2.630
- e) None of these

[2]

Q.17) A person, aged exactly 50, buys a whole life assurance with a sum assured of INR 75,000 payable at the end of the year of death. Calculate the standard deviation of the present value of this benefit using AM92 Ultimate mortality and 6% pa interest. :

- a) 11009
- b) 12089
- c) 15010

- d) 11502
- e) None of these

[2]

For Q18 and Q19 - On 1 January 2020, a life insurance company issued a number of without profit endowment policies maturing at age 60 to lives then aged 40 exact. The sum assured is payable at the end of year of death or on survival to the end of the term and level premiums are payable annually in advance throughout the term of the contract.

Premiums and reserves on each policy are both calculated on the following basis:

Mortality AM92 Select

Interest 4% per annum

Initial commission 60% of the first premium

Renewal commission 6% of each annual premium excluding the first

Q.18) Calculate the annual office premium per 1,000 Sum Assured for each policy.

- a) 35.45
- b) 36.32
- c) 36.99
- d) 36.98
- e) None of these

[2]

Q.19) Calculate the gross premium prospective reserve per 1,000 sum assured for each policy in force at 31 December 2024.

- a) 164.2
- b) 166.0
- c) 166.3
- d) 164.5
- e) None of these

[2]

Q.20) Under a 20-year policy issued by a life insurance company, the benefit payable on death, at the end of the year of death, is a return of total premiums paid without interest.

A premium of Rs. 631 is payable annually in advance, throughout the term of the policy.

The following information is available for a policy in force at the start of the 19th year:

Reserves at the start of the 19th year: 17,095

Reserves at the end of the year per survivor :18,510

Probability of death during the year: 0.015

Rate of interest earned: 4.5% per annum

Determine the profit or loss which is expected to emerge at the end of the 19th year for each policy in force at the start of that year. Ignore expenses and all decrements other than death.

- a) 111.5
- b) -166.2
- c) 121.0
- d) -156.7
- e) None of these

[2]

For Q21 and Q22: A life insurance company issues 4-year unit-linked endowment assurance policies to lives aged 50 exact. Level premiums are payable annually in advance until maturity or earlier death.

The company has performed a profit test on these policies and the profit vector per policy sold, ignoring surrenders, is as follows: (170, -120, -7, 15)

Mortality: AM92 Ultimate

Interest on non-unit cash flows: 4% per annum

Ignore expenses

Q.21) Calculate the profit signature per policy sold

- a) Year 1: -43.34 Year 2: 0 Year 3: 0 Year 4: 13
- b) Year 1: 170 Year 2: 0 Year 3: 0 Year 4: 14
- c) Year 1: 170 Year 2: -119.7 Year 3: -6.96 Year 4: 14.87
- d) Year 1: 169.57 Year 2: -119.36 Year 3: -6.94 Year 4: 14.82
- e) None of these

[2]

Q.22) Calculate the profit signature per policy sold at time 1 if negative non-unit fund cash flows are zeroized.

- a) 50.02
- b) 48.01
- c) 45.23
- d) 45.98
- e) None of these

[2]

Q.23) An individual aged 25 exact wishes to purchase a property at age 47, which is currently priced at INR 50,00,000. He plans to save a fixed amount, X, monthly in advance during this term.

The investment return is expected to be 2.5% p.a. effective, and property price inflation is expected to be 0.5% effective per half year.

Calculate X, using annuity functions.

- a) 14,191.42

- b) 15,895.33
- c) 17,739.30
- d) 24,521.92
- e) None of these

[2]

Q.24) Explain $(\ddot{I}a)_{20}$

- a) Present value of an increasing annuity-arrears payable for 20 time units, the i th payment (of amount t) being made at time t .
- b) Present value of an annuity-due payable for 20 time units, the i th payment (of amount t) being made at time $t-1$.
- c) Present value of an increasing annuity-due payable for 20 time units, the i th payment (of amount t) being made at time $t-1$.
- d) Present value of an annuity-arrears payable for 20 time units, the i th payment (of amount t) being made at time t .
- e) None of these.

[2]

For Q25, Q26 and Q27: A three-state transition model is as below:

Healthy to ill = σx

ill to Healthy = ρx

Healthy to Dead = μx

ill to Dead = γx

Assume that the transition probabilities are constant at all ages with $\sigma x = 3\%$, $\gamma x = 6\%$, $\rho x = 1\%$ and $\mu x = 4\%$. Given constant force of interest of δ pa. select the correct integral expression for the expected present value of each of the following benefits payable to a policyholder aged 45.

Q.25) A healthy life aged 45 years exact takes out a 20-year sickness contract that provides a premium refund of Rs.35,000 if the insured remains Healthy for the full duration of the contract.

- a) $35000 * e^{-\delta t} * \int_0^{20} e^{-(0.03+0.04)} dt$
- b) $35000 * \int_0^{20} e^{-(0.03+0.04)} * e^{-\delta t} dt$
- c) $35000 * \int_0^{20} e^{-(0.03+0.01)} * e^{-\delta t} dt$
- d) $35000 * \int_0^{20} e^{-(0.03+0.01)} dt$
- e) None of these

[2]

Q.26) Benefit of 5,00,000 payable immediately on death, provided that death occurs within the next 20 years:

- a) $500000 * \int_0^{20} (0.04 * tP45(HH) + tP45(HI) * 0.06) * e^{-\delta t} dt$
- b) $500000 * \int_0^{20} (0.04 * tQ45(HH) + tP45(HI) * 0.06) * e^{-\delta t} dt$

- c) $500000 * \int_0^{20} (0.04 * tP45(HH) + tQ45(HI) * 0.06) * e^{-\delta t} dt$
 d) $500000 * \int_0^{20} (0.04 * tP45(HH) + tP45(HI) * 0.06) * e^{\delta t} dt$
 e) None of these

[2]

Q.27) Benefit of 1,00,000 payable continuously during sickness for maximum 20 years:

- a) $100000 * \int_0^{20} tP45(HH) * 0.04 * \overline{a}_{20} e^{-\delta t} dt$
 b) $100000 * \int_0^{20} tP45(HH) * 0.04 * e^{-\delta t} dt$
 c) $100000 * \int_0^{20} tP45(HI) * \overline{a}_{20} e^{-\delta t} dt$
 d) $100000 * \int_0^{20} tP45(HI) * e^{-\delta t} dt$
 e) None of these

[2]

Q.28) An investor pays INR 400 every half-year in advance into a 25-year savings plan. Calculate the accumulated fund at the end of the term if the interest rate is 6% per annum convertible monthly for the first 15 years and 6% per annum convertible half yearly for the final 10 years. Do not round off at intermediate stages

- a) 45,330
 b) 30,780
 c) 46,703
 d) 46,965
 e) None of these

[4]

Q.29) A fund had a value of INR 20,000 on 1st Jan 2020. A net cash flow of INR 5,000 was received on 1st Jan 2021 and a further net cash flow of INR 9,000 was received on 1st Jan 2022. Immediately before receipt of the first net cash flow on 1st Jan 2021, the fund had a value of INR 23,000, and immediately before receipt of the second net cash flow on 1st Jan 2022 the fund had a value of INR 31,000. The value of the fund on 1st Jan 2023 was INR 36,500.

Calculate the annual effective money weighted rate of return earned on the fund over the period 1st Jan 2020 to 1st Jan 2023

- a) 3.5%
 b) 3.1%
 c) 7.9%
 d) 7.2%
 e) None of these

[3]

Q.30) An insurance company has liabilities of INR 70,500 due in 6 years' time and INR 300,500 due in 15 years' time. The current interest rate is 8% per annum effective. Calculate the discounted mean term of liabilities.

- a) 12 years

- b) 12.13 years
- c) 12.05 years
- d) 12.10 years
- e) None of these

[2]

Q.31) A real estate development firm is evaluating a residential township project. The project is expected to generate revenues in three years after completion of the project.

If the market is favorable, the revenues will be Rs. 50 crores per year. If the market is not favorable, revenues will drop to Rs. 25 crores per year. The probability of a favorable market is $\frac{3}{4}$.

The initial cost of the project is Rs. 60 crores. If all regulatory approvals are received on time, the project will be completed within one year. Delays in approval will push the completion into year 2 and the project will be complete by the end of year 2. An additional cost of Rs. 20 crores will be incurred in Year 2 due to this delay. The probability of delay is $\frac{1}{3}$.

Assume all cash-flows occur at the start of each year. Calculate the expected NPV of the project at a risk discount rate of 8% p.a.

- a) INR 44.25 crores
- b) INR 43.79 crores
- c) INR 46.58 crores
- d) INR 51.05 crores
- e) None of these

[4]

Q.32) An investor purchases a bond 6 months after issue. The bond will be redeemed at 110% ten years after issue and pays coupons of 4% pa annually in arrears. The investor pays tax of 25% on income and 10% on capital gains. The purchase price of the bond per £100 nominal to provide the investor with a rate of return of 5% pa effective is denoted as:

- a) $P = 1.05^{\frac{1}{2}} * 3a_{10} + 99 v^{9.5} + 0.10Pv^{9.5}$
- b) $P = 1.05^{\frac{1}{2}} * 3a_{10} + 110 v^{9.5}$
- c) $P = 3a_{10} + 99 v^{9.5} + 0.10Pv^{9.5}$
- d) $P = 1.05^{\frac{1}{2}} * 3a_{10} + 99 v^{9.5}$
- e) None of these

[3]

Q.33) In January 2020, the government of a country issued an index-linked bond with a term of three years. Coupons were payable half-yearly in arrear, and the annual nominal coupon rate was 5% for per 100. Interest and capital payments were indexed by reference to the value of an inflation index with a time lag of six months. The inflation index was as follows:

Inflation date	Index
Jul-19	107.3
Jan-20	113.5
Jul-20	120.0

Jan-21	126.4
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Compute the coupon payable for Jan21

- a) 3.08
- b) 2.64
- c) 5.59
- d) 2.80
- e) None of these

[2]

Q.34) A special whole life assurance policy is issued to a life aged 45 exact. The policy provides a benefit of INR 40,000 on death within 15 years of inception, and INR 50,000 on death thereafter. Benefits are payable at the end of the year of death. Calculate the expected present value of the benefit payments.

Basis:

Mortality: AM92 Ultimate

Interest: 6% p.a. effective

- a) 23845.26
- b) 8027.86
- c) 7670.11
- d) 32511.02
- e) None of these

[2]

Q.35) A life insurance company issues a joint life annuity policy to two lives male aged 65 and female aged 63 exact in return for a single premium. Under the policy, an annuity of INR 30,000 per annum is payable annually in advance while at least one of the lives is alive. Calculate Single Premium under the policy.

Basis:

Mortality: PMA92C20 for the first life, PFA92C20 for the second life

Interest: 4% per annum

Expenses: Ignored

- a) 504,420
- b) 475,770
- c) 511,920
- d) 509,700
- e) None of these

[3]

Q.36) A life insurance company issued a with profits whole life policy to a life aged 30 exact, on 1st Jan 2022. Under the policy, the basic sum assured of INR 10,00,000 and attaching bonuses are payable immediately on death. The company declares simple reversionary bonuses of 1% per annum at the start of each year.

Level premiums are payable annually in advance under the policy.

Basis:

Mortality: AM92 Ultimate

Interest: 6% per annum

Expenses: Initial Expense: INR 500

No subsequent expense

Assume bonus entitlement earned immediately on payment of premium.

Calculate the annual premium, using the equivalence principle.

- a) 6,402.47
- b) 6,249.19
- c) 6,429.10
- d) 6,492.13
- e) None of these

[3]

Q.37) On 31st Dec 2024, the policy (as stated in Q36 above) is still in force. A total of INR 40,000 has been declared as a simple bonus to date on the policy. The company calculates provisions for the policy using a gross premium prospective basis, with the following assumptions:

Mortality: AM92 Ultimate

Interest: 4%

Bonus loading: 1.5% per annum simple

Calculate the provision for the policy as at 31st Dec 2024. Kindly use premium calculated above upto 2 decimal only.

- a) 161,817
- b) 163,176
- c) 172,898
- d) 155,803
- e) None of these

[3]

Q.38) Consider the following group of whole life assurance policies:

- Year of Issue: 2019
- Number in force at the policy anniversary in 2024: 1,900
- Number in force at the policy anniversary in 2025: 1,867

- Exact age at the policy anniversary in 2024: 70
Sum Assured: 60,000 per policy, payable immediately on death
- Level premiums are payable annually in advance for the whole life.
- Mortality : AM92 Ultimate
- Interest rate: 4% p.a

Calculate the mortality profit for this group of policies for the policy year commencing at the policy anniversary in 2024, assuming death is the only cause of policy termination, and that the insurer holds net premium reserves for these contracts calculated assuming AM92 Ultimate mortality.

Where, net premium = 2,631.05 ; ${}_6V = 11,354.90$; $q_{70} = 0.024783$; $q_{71} = 0.027606$

- 702,037
- 741,249
- 969,328
- 1,008,540
- None of these

[4]

For Q39 and Q40. A life insurance company issues a 4-year unit-linked endowment assurance contract to a male life aged 45 exact under which level premiums of INR 100,000 per annum are payable in advance.

In the first year, 95% of the premium is allocated to units and 102.5% in the second and subsequent years. The units are subject to a bid-offer spread of 5% and an annual management charge of 0.5% of the bid value of the units is deducted at the end of each year.

If the policyholder dies during the term of the policy, the death benefit of INR 500,000 or the bid value of the units after the deduction of the management charge, whichever is higher, is payable at the end of the year of death.

The company uses the following assumptions in its profit test of this contract:

- Rate of growth on assets in the unit fund: 6% per annum
- Rate of interest on non-unit fund cashflows: 4.3% per annum
- Independent rates of mortality AM92
- Risk discount rate 8% per annum
- Initial expenses INR 2500
- Renewal expenses INR 500 per annum on second & subsequent premium dates
- Surrenders: None
- $q_{45} = 0.001465$ $q_{46} = 0.001622$ $q_{47} = 0.001802$ $q_{48} = 0.002008$

Q.39) Calculate and select the correct value of projected Unit Fund at the end of year 1, year 2, year 3 and year 4 from the following:

- 95187; 203095; 316905; 436942
- 96983; 208857; 329077; 458265

- c) 93660; 198253; 306799; 419445
- d) 95187; 203095; 316905; 406942
- e) None of these

[4]

Q.40) Calculate and select the correct value of projected net cash flows (non unit fund) at the end of each year of year 1, year 2, year 3 and year 4 from the following:

- a) 7447; 2755; 3479; 4225
- b) 7450; 2774; 3531; 4393
- c) 7445; 2739; 3435; 4197
- d) 7447; 2755; 3479; 4285
- e) None of these

[4]

Q.41) A life insurance company offers a 10-year endowment assurance products, with an initial sum assured of INR 50,000, to lives aged 55 exact. Level premiums are payable annually in advance throughout the term of the policy or until earlier death.

The benefits are payable at the end of the year of death, or on survival to the end of the term.

A with-profit endowment assurance with an expected simple reversionary bonus of 2.0% p.a. added at the end of each year. Each increase is applied at the end of the year if the policyholder is still alive. Calculate the annual premium. Do not round off any intermediate level.

Basis:

Mortality: AM92 Ultimate

Interest: 6% p.a effective

- a) 6430.43
- b) 4451.98
- c) 4724.07
- d) 4696.86
- e) None of these

[4]
